



In silico genome-wide analysis of *Citrus sinensis* (L.) Osbeck *NHX* and *KEA* genes and their roles in abiotic stress

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Abstract

Citrus sinensis (L.) Osbeck is a valuable plant belonging to the Rutaceae family. Stress is induced in plants by abiotic factors such as drought, salinity, and temperature. The monovalent cation proton antiporter (CPA) superfamily, which includes the *K⁺ efflux antiporter (KEA)* and *Na⁺/H⁺ exchanger (NHX)* genes, plays a crucial role in the regulation of physiological events. This study aims to identify the *KEA* and *NHX* genes of *C. sinensis* and elucidate the roles of these genes in the response to abiotic stress. For this purpose, phylogenetic structure, distribution of chromosomes, gene duplications, gene and protein structures, cis-acting elements, functional gene ontologies, targeted miRNAs, and in silico PCR primer searches were performed using *CsNHX* and *CsKEA* sequences. Two *KEA* and fifty-five *NHX* were identified as a result of the analysis. Nine of the fifty-five genes (*CsNHX5*, *CsNHX11*, *CsNHX12*, *CsNHX17*, *CsNHX27*, *CsNHX28*, *CsNHX47*, *CsNHX48*, and *CsNHX55*) have been identified as playing a role in the stress response. On the phylogenetic tree, *NHX* genes were observed to be divided into three distinct clusters. The existence of multiple segmental and tandem duplications in the *CsNHX* genes has been demonstrated. Stress-related motifs were identified in the promoter regions of *CsKEA* and *CsNHX* by cis-acting element analysis, while stress-related miRNAs were identified by miRNA analysis. Consequently, *KEA* genes are responsible for transport, but they may also play a role in abiotic stress, as they contain cis-acting elements involved in the stress response and are targeted by miRNAs associated with stress. In addition, it has been determined that *CsNHX5*, which plays a role in the stress response, has the potential to be used in future transgenic plant production studies as it satisfies the PCR in silico criteria.

Keywords miRNA · Protein structure · PCR · Rutaceae · Stress response

1 Introduction

To maintain their normal growth and development in the face of abiotic stresses such as salinity, temperature, and drought, plants develop numerous protective mechanisms (Xiaolin et al. 2022). Ion homeostasis is crucial to the life cycle of plants and their resistance to adverse environmental conditions (Isayenkov et al. 2020). CPA superfamily controls physiological events such as Na^+ , K^+ , and pH homeostasis, which are important for plant life (Kong et al. 2021). In addition, CPA genes also play essential roles in plant growth, development, signal transduction, and stress responses (Hus-sain et al. 2022). The CPA superfamily is divided into CPA1

and CPA2 groups. CPA1 is composed of the *NHX*, whereas CPA2 is composed of the *KEA* and *cation/H⁺ exchanger (CHX)* families (Sharma et al. 2020). Numerous physiological processes related to vacuolar ion homeostasis are regulated by NHXs (Bassil et al. 2019). NHX antiporters play a role in growth and development-related events such as stomatal conductance, membrane trafficking, pH regulation, cell expansion, flower development, protein processing, and the salt stress response (Zhou et al. 2016). According to their location, these antiporters can be classified as cell membrane-NHXs, vacuolar-NHXs, or endosomal-NHXs (Khare et al. 2021). Plants employ various strategies to mitigate the damage caused by high salinity, such as compartmentalization, exclusion, and secretion of Na^+/H^+ ions into vacuoles. NHX proteins, particularly under stressful conditions, assist in sequestering Na^+ ions into vacuoles, thereby reducing the toxic Na^+ concentration in the cytoplasm. These NHX proteins within vacuoles regulate the exchange of Na^+/H^+

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across the tonoplast membrane, thus mitigating the harmful effects of excessive Na^+ levels and preserving cellular turgor while safeguarding essential enzymatic processes in the cytoplasm (Yarra 2019). The overexpression of *NHX* genes may have an effect on the upregulation of genes related to stress resistance, potentially contributing to responses to salt stress. It is noted that *AtNHX5–6* can regulate K^+ and pH homeostasis under saline conditions and enhance salt and K^+ resistance (Ma et al. 2020).

Phylogenetic analysis categorizes KEA proteins into two clades: KEAI and KEAII. KEAI, based on its N-terminal characteristics, is further subdivided into two groups: KEAIa, featuring an extended soluble N-terminal domain spanning approximately 570–770 amino acids, and KEAIb, which lacks the N-terminal extension (Isayenkov et al. 2020). On a subcellular level, the KEA protein group plays a crucial role in potassium homeostasis. Within Arabidopsis, these KEA proteins can be categorized into two subgroups: KEA1–3, which are primarily associated with plastids, and KEA4–6, which demonstrate a closer connection to the endomembrane system (Imtiaz et al. 2023). According to research (Rehman et al. 2017), the *KEA* gene family is important for the K^+ balance and is linked to abiotic stresses such as excessive salt and drought. In soybean, it is noted that most of the *KEA* genes respond to K^+ deficiency and salinity, and similarly, in *Triticum aestivum* L., the *KEA* gene expression profile may function in response to stress (Isayenkov et al. 2020).

Citrus sinensis L., is a fruit known as the "sweet orange," belonging to the Rutaceae family, distributed in tropical and subtropical regions (Liew et al. 2018). *C. sinensis* contains alkaloids, flavonoides, steroids, hydroxyamides, alkanes, fatty acids, volatile compounds, carotenoids, and minerals such as potassium, magnesium, calcium, and sodium. In addition to being a high source of vitamin C, it has been used traditionally to cure colds, coughs, bronchitis, obesity, constipation, diarrhea, menstruation disturbances, cardiovascular disorders, stress, and depression (Mannucci et al. 2018). It is essential to incorporate *C. sinensis* into a daily diet. In the past few years, people have tried to find effective treatments for COVID-19, a disease that has killed a lot of people, and research has been conducted on this subject. One of the studies on this topic (Mangukia et al. 2022) revealed a strong interaction between the *csi-miR169–3p* and the conserved region "CTGCCT" found in the reference SARS-CoV-2 genome and all 770 curated Omicron variants from 37 regions worldwide. This region is believed to be conserved in other variants of SARS-CoV-2 as well, suggesting that the consumption of plants like *C. sinensis* may play a role in enhancing people's immunity against this disease.

Bioinformatics plays a crucial role in handling and interpreting data generated by omics technologies, enabling scientists to filter valuable information, integrate data, and

support comprehensive views of system functionality. It provides accessible resources for pioneering research and is essential for the success and evolution of omics technologies. However, effective data utilization depends on scientists' awareness of data sources, technology sensitivity, and result quality, especially for inexperienced users seeking novel insights into biological systems (Ambrosino et al. 2020). *1-amino-cyclopropane-1-carboxylic acid synthase* (ACS) (Sun et al. 2022), *polyamine oxidase* (PAO) (Wang and Liu 2015), *laccase* (LAC) (Xu et al. 2019), *auxin response factor* (ARF) (Lin et al. 2015), *histone modification* (Xu et al. 2015), *R2R3-MYB transcription factor* (Liu et al. 2014), *glyceraldehyde-3-phosphate dehydrogenase* (GAPDH) (Miao et al. 2019), *WUSCHEL-related homeobox* (WOX) (Shafique Khan et al. 2021), *Golden2-like* (GLK) (Xiong et al. 2022), *GRAS* (Zhang et al. 2019), *respiratory burst oxidase homology* (*Rboh*) (Zhang et al. 2022a, b) gene families and pectin methylesterase inhibitors (PMEI) (Li et al. 2022), aquaporin genes (Wei et al. 2019), major intrinsic proteins (de Paula Santos Martins et al. 2015) etc. analyzes were made in *C. sinensis*. On the other hand, there is no research on genome-wide analysis for the *NHX* and *KEA* genes of *C. sinensis*.

The primary objective of this study is to conduct an extensive genome-wide investigation into the *KEA* and *NHX* gene families in *C. sinensis*, employing advanced bioinformatics tools. By unraveling the intricate roles of these gene families in the context of abiotic stress, our research aims to offer a comprehensive resource that not only advances our comprehension of *C. sinensis* but also paves the way for innovative solutions to enhance crop resilience in the face of adverse environmental factors.

2 Materials and methods

Identification of *NHX* and *KEA* genes in *C. sinensis* – The *NHX* and *KEA* protein sequences were retrieved from the NCBI (National Center for Biotechnology Information) database (<https://www.ncbi.nlm.nih.gov/>, accessed on May 31, 2022). Using the BLASTP tool of QIAGEN CLC Genomics Workbench 22.0.1, these sequences and *Citrus sinensis* protein sequences were compared. In addition, the conserved region analysis of *NHX* and *KEA* proteins was carried out in the PFAM 35.0 database (<https://pfam.xfam.org/>, accessed on May 31, 2022) using the QIAGEN CLC Genomics Workbench 22.0.1. After performing a comparison, repetitive sequences were eliminated, and potential *NHX* and *KEA* proteins were identified. The physicochemical parameters of the identified proteins were then computed using the ExPASy ProtParam program (<https://web.expasy.org/protparam/>, on May 31, 2022).

Phylogenetic analysis – The amino acid sequences were imported into the MEGA 11 program, and MUSCLE was used to align multiple sequences. The aligned file was used to generate a phylogenetic tree using the Maximum Likelihood Tree technique with bootstrap analysis for 1000 replicates. The Jones–Taylor–Thornton (JTT) substitution model was used to reconstruct the phylogeny. Following this procedure, the data were exported in Newick format and submitted to the Interactive Tree Of Life (iTOL) v6 (<https://itol.embl.de/>, viewed on September 7, 2022). This online software program was used to visualize the phylogenetic tree.

Chromosome distribution – The genomic sequences of the *KEA* and *NHX* proteins were examined in the NCBI database to find the chromosomal localizations of the genes. The MG2C (MapGene2Chromosome) v2.1 software (http://mg2c.iask.in/mg2c_v2.1/, accessed on September 23, 2022) was utilized to visualize chromosomal locations.

Determination of *KEA* and *NHX* orthologous genes and estimation of divergence times – Tandem and segmental duplications were detected for the *KEA* and *NHX* genes, and a comparison was made between the duplications of these genes and orthologous genes in the genome of *Citrus clementina* Hort. Blastp scanning was conducted at NCBI, and values with an expected value of $\leq 1e-50$ among sequences displaying similarity and matching 50% similarity parameters were used as data. The Clustal Omega online tool (<https://www.ebi.ac.uk/Tools/msa/clustalo/>, accessed on July 2, 2022) was used to align tandemly and segmentally duplicate the *KEA* and *NHX* genes and their orthologs in *C. clementina*. The synonymous (Ks) and non-synonymous (Ka) substitution rates with protein sequences and coding sequences (cDNA) of these genes were computed using PAL2NAL (<http://www.bork.embl.de/pal2nal/>, accessed on July 2, 2022). The evolutionary divergence times of orthologous gene pairs were calculated using the equation $T = (\text{Mya, Million years ago}) = Ks/2\lambda$ ($\lambda = 6.5 \times 10^{-9}$) (Arslan et al. 2021).

Analysis of gene structure and conserved protein motif – Exon–intron regions were identified by comparing the genomic sequences with the predicted coding sequences (CDS) in the Gene Structure Display Server (GSDS 2.0, <http://gsds.gao-lab.org/>, accessed on June 23, 2022). *KEA* and *NHX* protein sequences were analyzed using the MEME (Multiple Em for Motif Elicitation) Suite 5.4.1 version online program (<https://meme-suite.org/meme/>, accessed on July 3, 2022). Then, MAST (Motif Alignment & Search Tool) was used to view the detailed information of the motifs. In the analysis, using classical mode, the maximum number of motifs was determined as 10, and the optimum width was set between 6 and 50.

Homology modeling and protein interaction of *KEA* and *NHX* – Homology modeling for *KEA* and *NHX* proteins was performed using intensive mode in the Phyre² (Protein Homology/analogy Recognition Engine V 2.0) program; <http://www.sbg.bio.ic.ac.uk/~phyre2/html/page.cgi?id=index>, viewed on June 26, 2022), and the protein interaction networks were predicted using STRING 11.5 (<https://string-db.org/>, retrieved on September 16, 2022).

Cis-acting elements analysis – To evaluate the *KEA* and *NHX* genes' cis-regulatory elements, the promoter sequences (2 Kb sequences upstream of start codons) were extracted from the NCBI database. The promoter sequences were scanned with the PlantCARE database (<https://bioinformatics.psb.ugent.be/webtools/plantcare/html/>, accessed on July 3, 2022). Moreover, the cis-acting elements were counted using the Microsoft Excel program.

Subcellular localization and functional gene ontology analysis – The CELLO2GO (<http://cello.life.nctu.edu.tw/cello2go/>, accessed on August 10, 2022) was utilized to estimate the subcellular localization of the *KEA* and *NHX* genes and to analysis gene ontology.

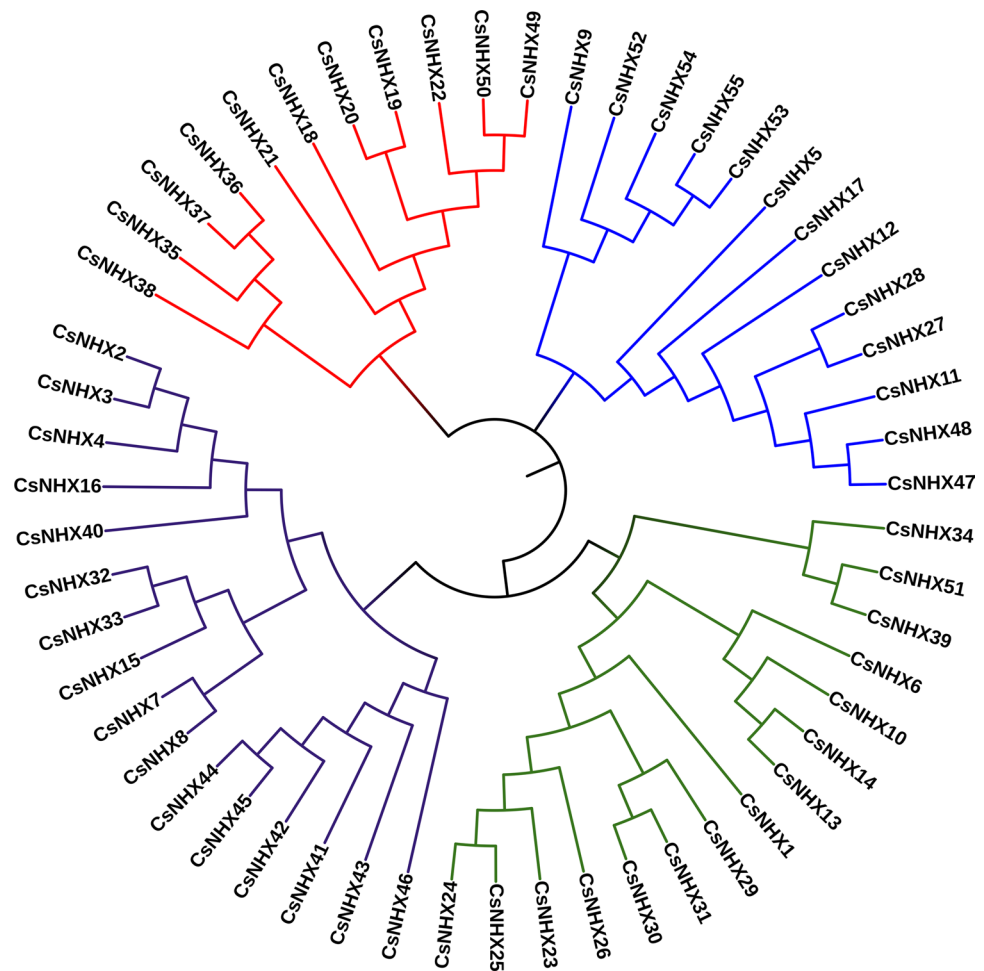
miRNA analysis – The coding sequences (CDS) of *KEAs* and *NHXs* were used to recognize potential target miRNAs within the psRNATarget database (<https://www.zhaolab.org/psRNATarget/>, accessed on June 23, 2022). The obtained results were transferred to Microsoft Excel, the miRNAs were determined in the microRNA database (miRBase) (<https://www.mirbase.org/>, retrieved on June 23, 2022), the plant miRNAs were selected, and these miRNAs were used in the article in excel format.

In silico PCR primer search – A search for in silico primers was conducted using the FastPCR Professional 6.8.04 program.

3 Results

Identification of *KEA* and *NHX* genes in *C. sinensis* – After comprehensive searches, two *KEA* genes and fifty-five *NHX* genes were identified in the *C. sinensis* genome. Based on the gene distribution information on the chromosomes, these genes were renamed. Gene names were created by using the initial letters of the plant's Latin name, followed by the digits *KEA* and *NHX*. When the molecular weights and amino acid lengths of the identified *KEA* genes were evaluated, the *C. sinensis* *KEA* gene molecular weights ranged from 86,308.34 Da to 130,248.64 Da, and their protein lengths ranged from 793 to 1207 aa. In terms of *NHX* genes, when we examined the molecular weights and protein lengths, it

Fig. 1 The phylogenetic tree of *NHX* genes (Cluster I, marked in red; Cluster II, blue; Cluster IIIa, violet; Cluster IIIb, green)



was determined that their molecular weights were between 11,733.25 Da and 127,142.80 Da, and the protein lengths varied between 109 and 1148 aa. The physicochemical investigation revealed that all CsKEA proteins were acidic ($pI < 7$), whereas CsNHX proteins were either acidic or basic. The estimated range of theoretical isoelectric points (pI s) for CsKEA and CsNHX was 4.95–5.27 and 5.30–9.75, respectively. In addition, the KEA proteins exhibited an instability index ranging from 34.84 to 42.86, whereas the NHX values ranged from 14.43 to 47.76 (Supplementary File 1).

Phylogenetic analysis – On the *C. sinensis* NHX protein phylogenetic tree, three distinct groups were found. Cluster I included 11 CsNHX proteins, whereas Cluster II had 13 CsNHX proteins. In addition, Cluster III had 16 proteins in the "a" branch and 15 proteins in the "b" branch (Fig. 1). Observations indicated that CsNHX21 and CsNHX38 in Cluster I, CsNHX5 and CsNHX9 in Cluster II, CsNHX40 and CsNHX46 in Cluster IIIa, and CsNHX1, CsNHX6, and CsNHX34 in Cluster IIIb were distinct from the others.

Chromosome distribution – *C. sinensis* contains nine chromosomes. *CsKEA1* is located on chromosome 8, while *CsKEA2* is a scaffold. *NHX* genes were detected in each chromosome of *C. sinensis*. On chromosome 8, only the *CsNHX46* gene is present. The majority of genes (*CsNHX34*, *CsNHX35*, *CsNHX38*, *CsNHX39*, *CsNHX40*, *CsNHX41*, *CsNHX42*, *CsNHX43*, *CsNHX44*, and *CsNHX45*) are on chromosome 7. The *CsNHX* genes detected in the *C. sinensis* genome showed a scattered arrangement on the chromosomes. Furthermore, all genes between *CsNHX51* and *CsNHX55* are scaffold genes (Fig. 2).

Determination of KEA and NHX orthologous genes and estimation of divergence times – The orthologous relationships of the *KEA* and *NHX* genes in *C. sinensis* were determined with the genome of *C. clementina*, and divergence times were also estimated. As a result of orthologous gene analysis, six pairs of *KEA* genes and eight hundred sixty-nine pairs of *NHX* genes were detected. The estimated separation time of the *KEA* genes was 67.32 Mya, and the *NHX* genes were determined as 186.96 Mya. *KEA*'s average Ka/Ks ratio was calculated to be 0.09, while *NHX*'s was 0.10.

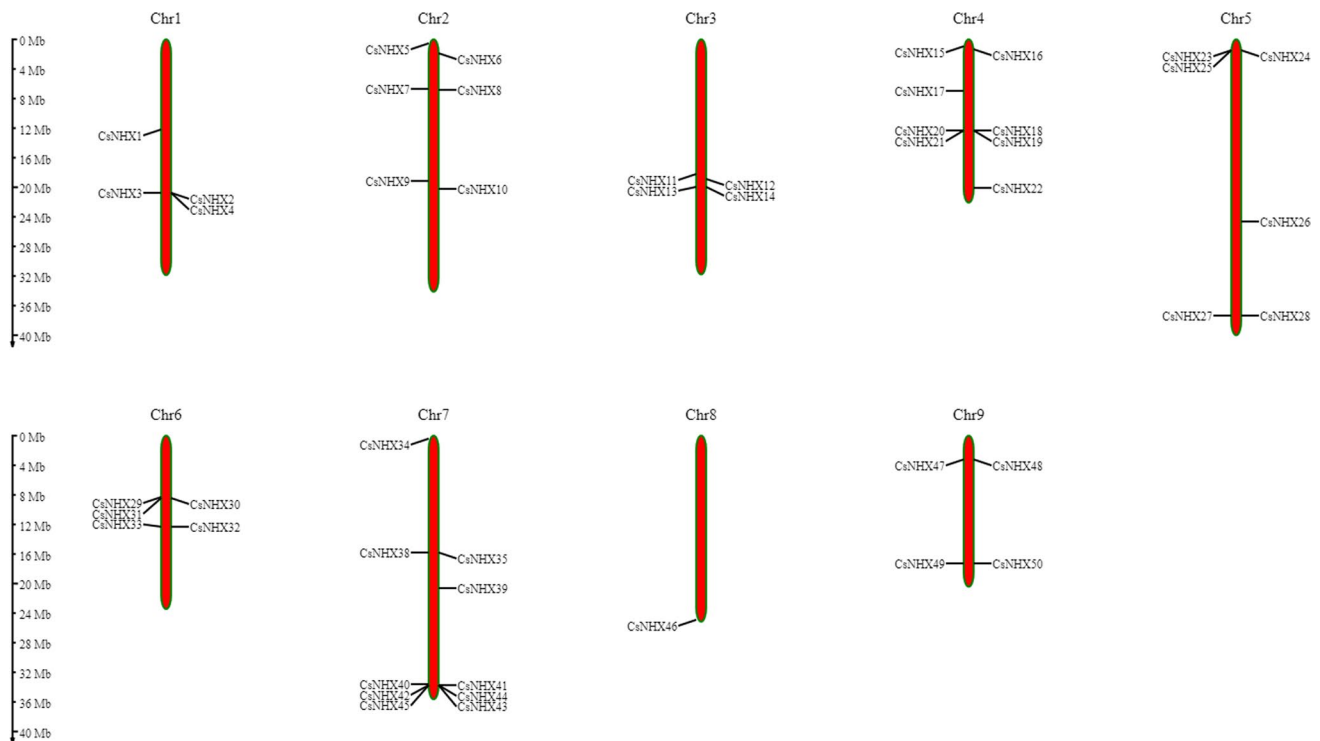


Fig. 2 Distribution of *NHX* genes on *C. sinensis* chromosomes

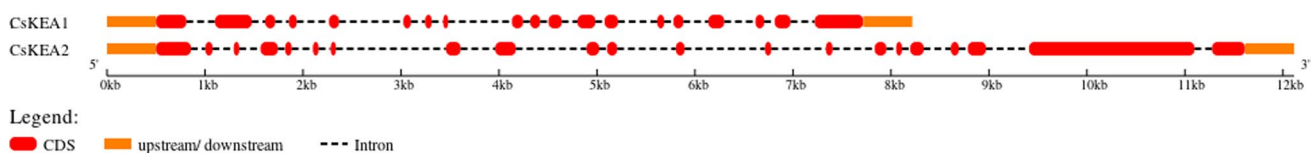


Fig. 3 Distribution of introns and exons in *CsKEA* genes

(Supplementary File 2). There were tandem and segmental duplications of *NHX* genes in *C. sinensis*; however, this analysis could not be conducted on the *KEA* genes since 1 of the 2 *KEA* genes in this plant is a scaffold gene. It was observed that forty-eight pairs of the *NHX* genes displayed tandem duplication and two hundred eighty pairs showed segmental duplication. The genes with the most tandem duplications were located on chromosome 7 (Supplementary File 3).

Analysis of gene structure and conserved protein motif – Revealing the gene structure is an important process that forms the basis of phylogenetic studies (Wei et al. 2021). *CsKEA* and *CsNHX* gene structure analyses and conserved motif analyses of the proteins of these genes were performed. *CsKEA1* contains 18 introns and 19 exons, whereas *CsKEA2* has 20 introns and 21 exons. Additionally, longer intron and exon regions were detected in *CsKEA2*. Moreover, *CsKEA1* and *CsKEA2* were detected to contain 5'-UTR and 3'-UTR (Fig. 3). In the article, the program was configured to have

a maximum of ten conserved motifs, and it was determined that both *KEA* proteins included each motif (Fig. 4).

The investigation of the *CsNHX* gene structure revealed that the *CsNHX4* gene has the shortest length and lacks introns. The genes with the fewest number of introns and exons are *CsNHX10*, *CsNHX32*, *CsNHX42*, and *CsNHX46* (1 intron, 2 exons). Nonetheless, a lengthy exon region has been found in the *CsNHX42* gene. The gene with the greatest number of introns and exons was *CsNHX54* (22 introns and 23 exons), whereas the *CsNHX52* gene contained 21 introns and 22 exons. In addition, 5'-UTR and 3'-UTR were identified in all genes (Fig. 5). Furthermore, phylogenetic tree and gene structure analysis support each other. Similar findings were achieved.

As a consequence of the conserved motif analysis of *CsNHX* proteins, it was determined that the majority of members of the same subgroup had the same motif composition. *CsNHX1*, *CsNHX2*, *CsNHX6*, *CsNHX14*, *CsNHX15*,

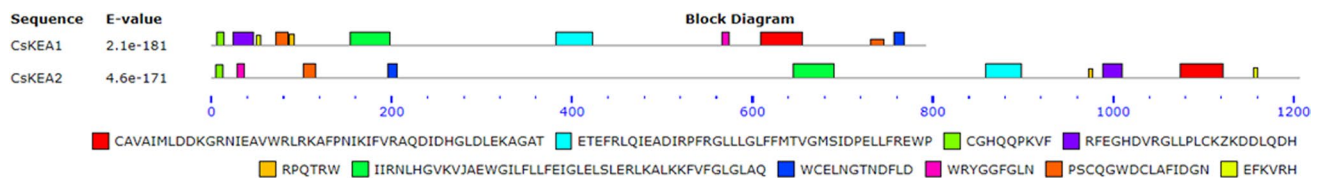


Fig. 4 Conserved motifs in CsKEA proteins

CsNHX23, CsNHX24, CsNHX25, CsNHX26, CsNHX29, CsNHX30, CsNHX31, CsNHX34, CsNHX41, CsNHX43, CsNHX44, and CsNHX45 proteins contain nine motifs, but only the 7th motif is absent in these proteins. In contrast, CsNHX11, CsNHX12, CsNHX17, CsNHX47, and CsNHX48 only include the 6th and 7th motifs. Additionally, the 4th and 10th motifs were identified in CsNHX4, and the 1st and 7th motifs were found in the CsNHX5 protein. Moreover, only the 5th motif was detected in CsNHX35, CsNHX36, CsNHX37, and CsNHX38, and only the 6th motif was observed in CsNHX21 (Fig. 6). In addition, in conserved motif analysis, the motifs of sequences with an E-value of less than 10 are given below; therefore, the CsNHX9 protein is excluded from Fig. 6.

Homology modeling and protein interaction of KEA and NHX – 83% of residues in CsKEA1 and 94% in CsKEA2 were modeled with > 90% confidence. It was observed that CsKEAs were composed of α helices, β sheets and long loops. Furthermore, it was revealed that CsKEA2 included more α helices while CsKEA1 contained more parallel β sheets (Fig. 7).

Examining CsNHX proteins revealed that the helix-loop-helix structure is prevalent. All proteins with the exception of CsNHX2, CsNHX3, CsNHX23, CsNHX24, CsNHX26, CsNHX32, CsNHX33, CsNHX40, CsNHX42, CsNHX45, CsNHX46, and CsNHX54 are composed only of α helix and loop. CsNHX2, CsNHX3, CsNHX33, and CsNHX42 contain parallel β sheets, long loops, and multiple α helices. On the other hand, antiparallel β sheets, α helices, and long loops were detected in CsNHX54. It was revealed that CsNHX32 has parallel β sheets, antiparallel β sheets, alpha helix structures, and lengthy loops. Also observed were β turns, α helix and long loops in CsNHX23, CsNHX24, CsNHX26, CsNHX45, and CsNHX46. CsNHX11 and CsNHX47, unlike the other proteins, are composed of α helices and loops, and the α helix structure of these proteins is modeled separately from the other structures. Similarly, CsNHX40 consists of parallel β sheets, α helix and long loops, and this protein modeling is observed in two separate parts when viewed in detail (Fig. 8).

As a consequence of the analysis, interactions between 15 proteins were identified (CsNHX9, CsNHX10, CsNHX12,

CsNHX14, CsNHX17, CsNHX23, CsNHX25, CsNHX26, CsNHX29, CsNHX30, CsNHX31, CsNHX46, CsNHX48, CsNHX51, and CsNHX55). With the exception of CsNHX9, CsNHX12, and CsNHX48, all other proteins interact with CsNHX55. Therefore, CsNHX55 occupies the central position (Fig. 9).

Cis-acting elements analysis – As a result of the cis-acting element analysis, 25 motifs in *CsKEA* and 64 motifs in *CsNHX* have their functions characterized and named. The *CsKEA1* promoter was observed to contain more cis-acting elements than the *CsKEA2* promoter. *CsKEA* has LTR (low temperature) and MBS (drought) motifs associated with stress. While LTR was observed in *CsKEA1*, MBS was found in *CsKEA2*. In addition, Box4, chs-CMA1a, G-box, G-Box, GATA-motif, I-box, L-box, LAMP-element, MRE and TCCC-motifs related to light response were detected in *KEA* genes. According to the analysis in *CsNHX*, the most cis-acting element was observed in *CsNHX8* (159), followed by *CsNHX5* (158), and the least in *CsNHX1* (72).

CsNHX genes have stress-related LTR, MBS, TC-rich repeats (defense and stress), and WUN-motifs (wound). The genes containing the LTR are *CsNHX2*, *CsNHX9*, *CsNHX10*, *CsNHX15*, *CsNHX17*, *CsNHX18*, *CsNHX19*, *CsNHX20*, *CsNHX22*, *CsNHX26*, *CsNHX29*, *CsNHX30*, *CsNHX31*, *CsNHX40*, *CsNHX43*, *CsNHX47*, and *CsNHX51*. MBS was identified in the promoter region of most genes (*CsNHX1*, *CsNHX4*, *CsNHX6*, *CsNHX8*, *CsNHX12*, *CsNHX13*, *CsNHX14*, *CsNHX17*, *CsNHX18*, *CsNHX22*, *CsNHX30*, *CsNHX32*, *CsNHX35*, *CsNHX36*, *CsNHX37*, *CsNHX38*, *CsNHX39*, *CsNHX40*, *CsNHX41*, *CsNHX43*, *CsNHX44*, *CsNHX46*, *CsNHX50*, *CsNHX51*). 20 genes included Tc-rich repeats (*CsNHX8*, *CsNHX11*, *CsNHX13*, *CsNHX14*, *CsNHX17*, *CsNHX26*, *CsNHX27*, *CsNHX28*, *CsNHX29*, *CsNHX31*, *CsNHX40*, *CsNHX41*, *CsNHX43*, *CsNHX44*, *CsNHX45*, *CsNHX46*, *CsNHX52*, *CsNHX53*, *CsNHX54*, *CsNHX55*). The WUN-motif has also been observed in many genes (*CsNHX2*, *CsNHX3*, *CsNHX4*, *CsNHX8*, *CsNHX9*, *CsNHX10*, *CsNHX15*, *CsNHX16*, *CsNHX18*, *CsNHX19*, *CsNHX20*, *CsNHX23*, *CsNHX26*, *CsNHX27*, *CsNHX28*, *CsNHX29*, *CsNHX35*, *CsNHX37*, *CsNHX38*, *CsNHX39*, *CsNHX40*, *CsNHX42*, *CsNHX43*, *CsNHX44*, *CsNHX45*, *CsNHX51*). Numerous motifs related to light response have

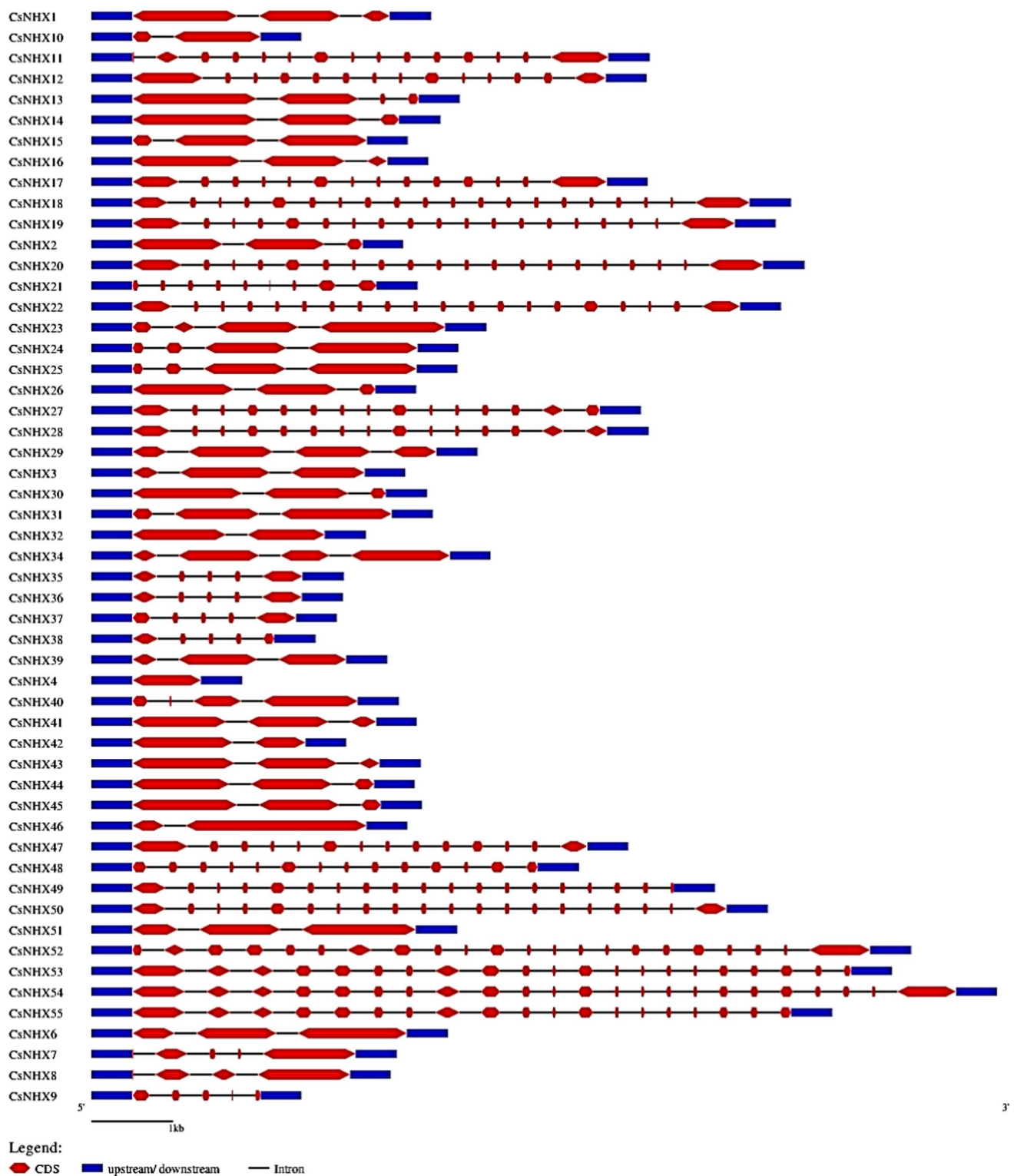


Fig. 5 Distribution of introns and exons in *CsNHX* genes

been identified in *CsNHX* promoter regions. These are listed as follows: 3-AF1 binding site, 4 cl-CMA1b, 4 cl-CMA2b, AAAC-motif, ACA-motif, ACE, AE-box, AT1-motif,

ATC-motif, ATCT-motif, Box II, Box 4, CAG-motif, chs-CMA1a, chs-CMA2a, chs-Unit 1 m1, G-box, G-Box, GA-motif, Gap-box, GATA-motif, GATT-motif, GT1-motif,

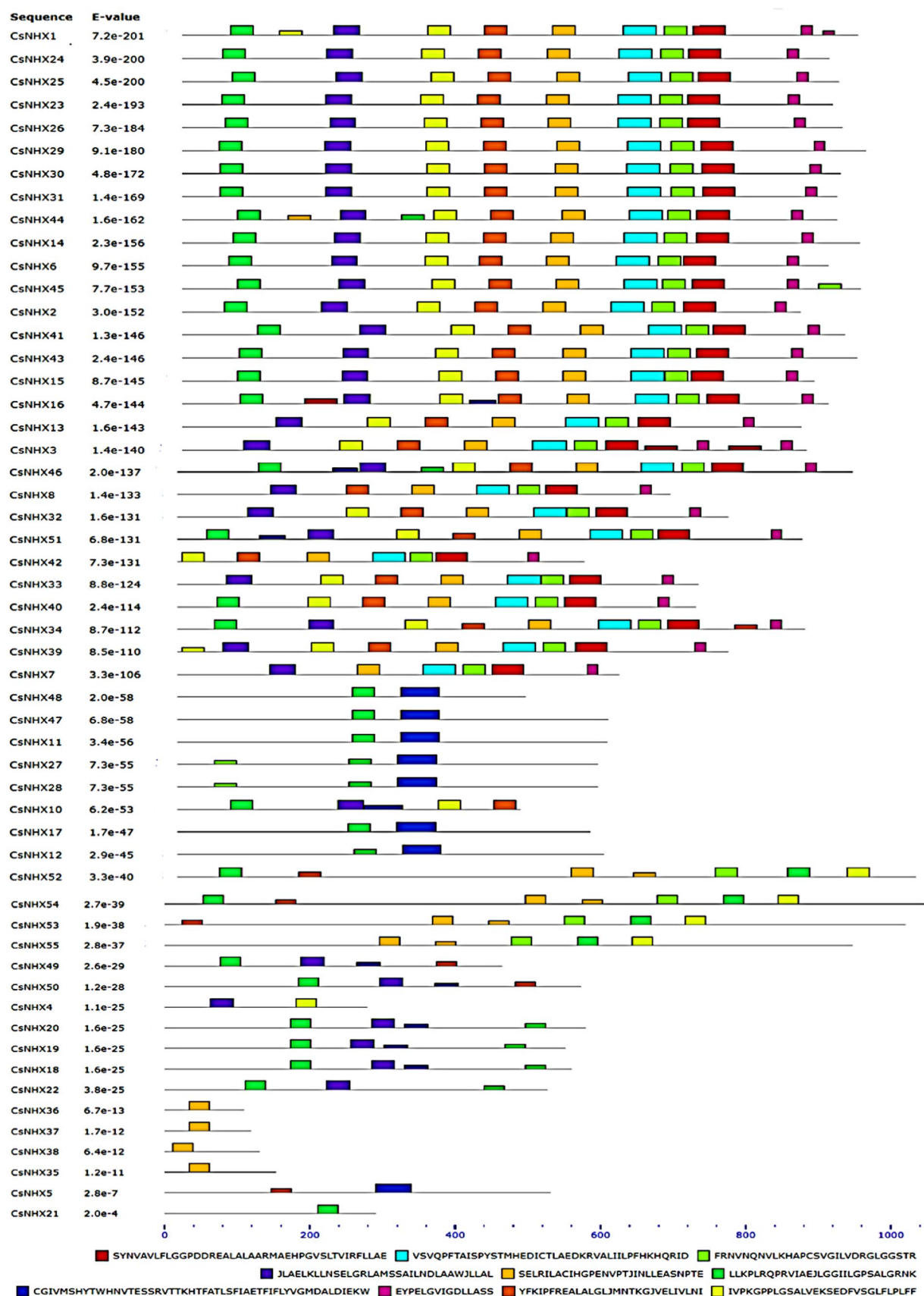


Fig. 6 Conserved motifs in CsNHX proteins

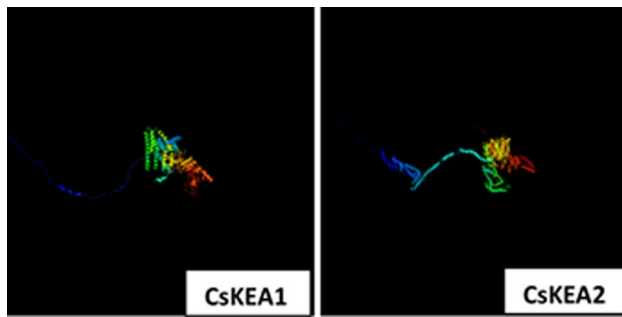


Fig. 7 The predicted three-dimensional structure of proteins belonging to the *CsKEA* genes

I-box, LAMP-element, MRE, Sp1, TCCC-motif, and TCT-motif (Supplementary File 4).

Phytohormones such as auxin, gibberellin, abscisic acid, ethylene, and jasmonic acid are involved in the process of resisting stress conditions (Manoli et al. 2018). *CsKEA* comprises the motifs ABRE (abscisic acid), CGTCA-motif (methyl jasmonate), GARE-motif (gibberellin), TGACG-motif (methyl jasmonate), TCA-element (salicylic acid), and SARE (salicylic acid). Besides the presence of these motifs in *CsNHX*, AuxRR-core (auxin), P-box (gibberellin), TATC-box (gibberellin), TGA-box (auxin) and TGA-element (auxin), are also seen. *CsNHX48* is the gene with the highest number of ABRE. AuxRR-core was determined only in *CsNHX26*, *CsNHX41*, *CsNHX49*, and *CsNHX50*. The CGTCA-motif and TGACG-motif were detected most number in *CsNHX11*. It was determined that P-box was found mostly in *CsNHX7*, while TCA-element was found mostly in *CsNHX51*. SARE was only identified in *CsNHX27* and *CsNHX28*. TATC-box was identified in *CsNHX2*, *CsNHX15*, *CsNHX23* (which is the most), *CsNHX35*, *CsNHX36*, *CsNHX37*, and *CsNHX38*. Additionally, the TGA-box was only found in *CsNHX22*. Also, the TGA-element was detected in *CsNHX2*, *CsNHX3*, *CsNHX10*, *CsNHX11*, *CsNHX13*, *CsNHX16*, *CsNHX40*, *CsNHX45*, *CsNHX49*, and *CsNHX50* (Supplementary File 4).

In terms of the analysis of cis-acting elements in general, it was determined that especially *CsNHX* genes were associated with important functions such as anaerobic induction, meristem expression, circadian control, anoxic specific inducibility, endosperm expression, differentiation of the palisade mesophyll cells, flavonoid biosynthetic gene regulation, cell cycle regulation, meristem specific activation, and seed-specific regulation. The GC-motif is the enhancer-like element involved in anoxic specific inducibility and is only observed in *CsNHX4* and *CsNHX51*. NON-box is the cis-acting regulatory element associated with meristem specific activation and is determined only in *CsNHX31*. RY-element is the cis-acting regulatory element implicated in seed-specific regulation, and

it is only detected in *CsNHX31*. Another important data is that TATA-box and CAAT-box were detected mostly in *CsNHX5* (Supplementary File 4).

Subcellular localization and functional gene ontology analysis

Cellular component, subcellular localization, biological process, and molecular function analyses of the *KEA* and *NHX* genes were predicted with the CELLO2GO program. The study revealed that *KEA* proteins are associated with transmembrane transporter activity and are localized in the plasma membrane. In addition, when examined as a cellular component, it was detected in intracellular, cell, cytoplasm, plasmid and organelle (Fig. 10).

NHX proteins have been identified in the extracellular and plasma membranes. In terms of molecular functions, it was observed that transmembrane transporter activity (58.48%) had the highest rate, followed by transferase activity and transferring acyl groups (28.18%). The biological processes of the *NHX* genes are very diverse. The ones with the highest rates of these functions are transport (32.05%), homeostatic process (18.72%), protein targeting (14.55%) and vacuolar transport (14.55%), respectively. The rate of other processes is lower. *NHX* genes also respond to stress (1.5%). *C. sinensis* has nine *NHX* genes (*CsNHX5*, *CsNHX11*, *CsNHX12*, *CsNHX17*, *CsNHX27*, *CsNHX28*, *CsNHX47*, *CsNHX48*, and *CsNHX55*) that are responsible for stress response. When *NHX* genes were analyzed in terms of cellular components, they were mostly observed in cells (12.98%), intracellular (12.98%), cytoplasm (12.98%), and organelles (12.98%) (Fig. 11).

miRNA analysis – *CsKEA* genes are targeted by miRNAs from 36 different plants (*Acacia auriculiformis* A.Cunn. ex Benth., *Arabidopsis lyrata* (L.) O'Kane & Al-Shehbaz, *Aquilegia coerulea* E.James, *Aegilops tauschii* Coss., *Arabidopsis thaliana* (L.) Heynh., *Bruguiera cylindrica* (Linnaeus) Blume, *Brachypodium distachyon* (L.) P.Beauv., *Bruguiera gymnorhiza* (L.) Lam., *Brassica oleracea* L., *Brassica napus* L., *Brassica rapa* L., *Cynara cardunculus* L., *Carica papaya* L., *C. sinensis*, *Hevea brasiliensis* Muell. Arg., *Hordeum vulgare* L., *Gossypium hirsutum* L., *Glycine max* (L.) Merr., *Gossypium raimondii* Ulbr., *Glycine soja* Siebold & Zucc., *Lotus japonicus* L., *Malus domestica* (Suckow) Borkh., *Medicago truncatula* Gaertn., *Nicotiana tabacum* L., *Oryza sativa* L., *Pinus densata* Mast., *Pinus taeda* L., *Populus trichocarpa* Torr. & A. Gray ex. Hook., *Rehmannia glutinosa* (Gaertn.) Steud., *Sorghum bicolor* (L.) Moench, *Solanum lycopersicum* L., *Solanum tuberosum* L., *Triticum aestivum*, *Theobroma cacao* L., *Vitis vinifera* L., *Zea mays* L.). The *CsKEA1* gene is targeted by 197 miRNAs and the *CsKEA2* by 260 miRNAs (Supplementary File 5).

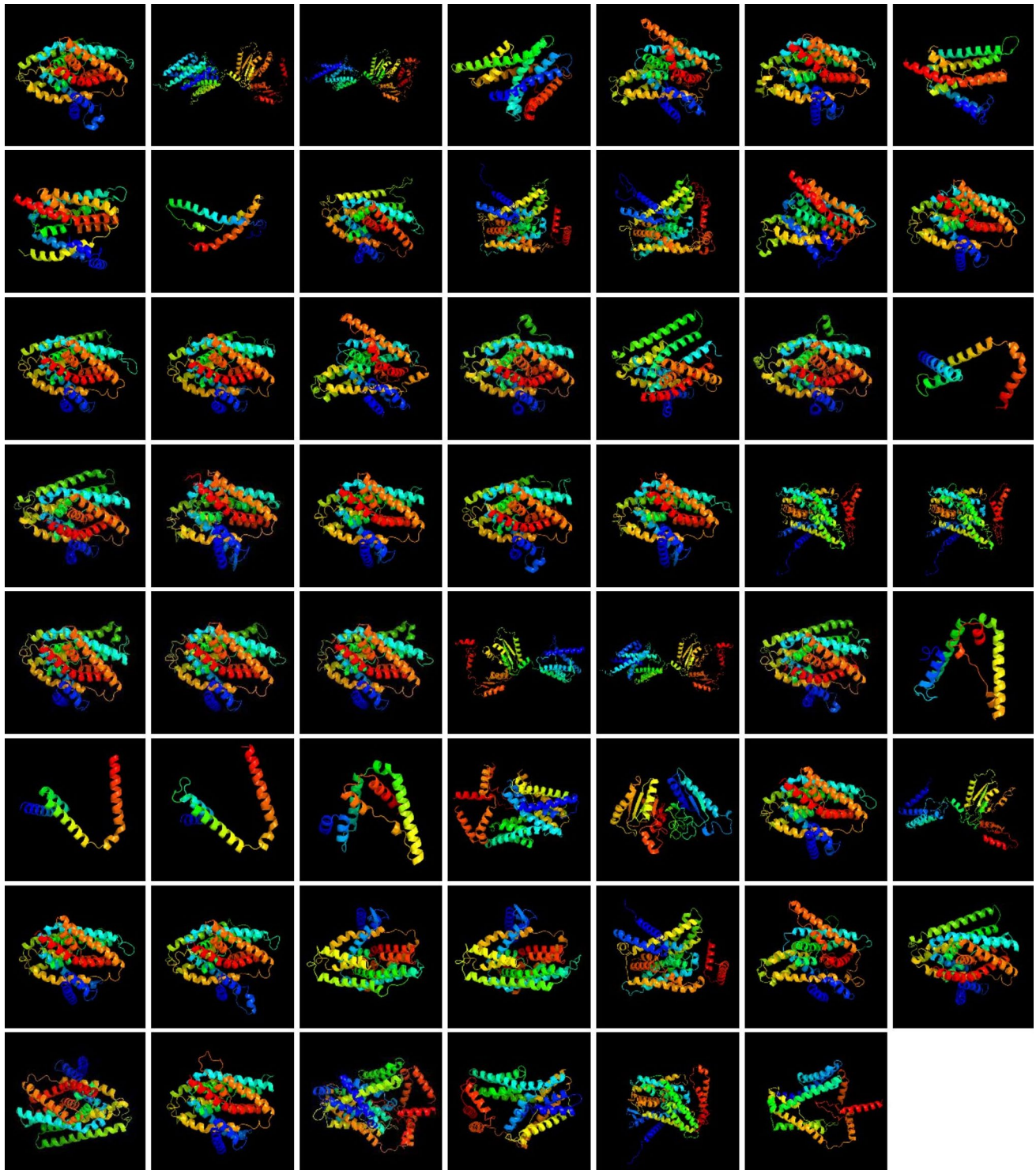


Fig. 8 The predicted three-dimensional structure of proteins belonging to the *CsNHX* genes (sorted by gene number)

CsNHX genes are targeted by miRNAs from 60 different plants (*Acacia auriculiformis*, *Arachis hypogaea* L., *Arabidopsis lyrata*, *Acacia mangium* Willd., *Aquilegia coerulea*, *Aegilops tauschii*, *Arabidopsis thaliana*, *Bruguiera cylindrica*, *Brachypodium distachyon*, *Bruguiera*

gymnorhiza, *Brassica napus*, *Brassica oleracea*, *Brassica rapa*, *Cynara cardunculus*, *Citrus clementina* Hort. Ex Tan, *Citrus reticulata* Blanco, *Citrus trifoliata* L., *C. sinensis*, *Cucumis melo* L., *Carica papaya*, *Digitalis purpurea* L., *Festuca arundinacea* Schreb., *Gossypium herbaceum* L.,

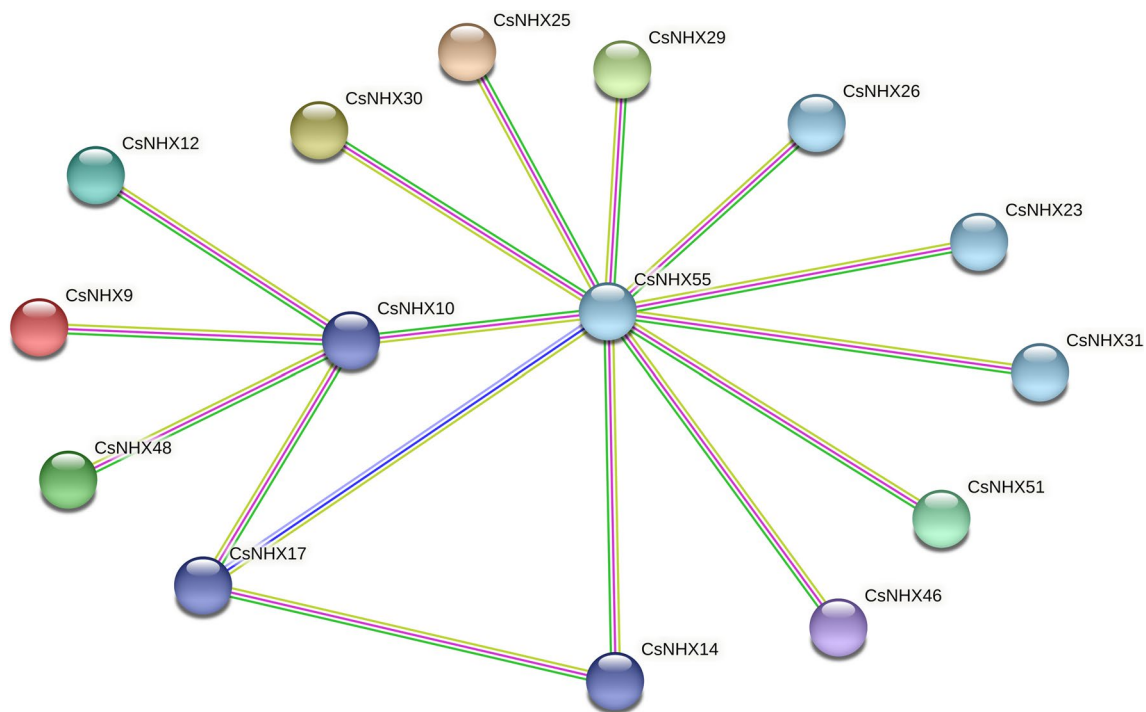
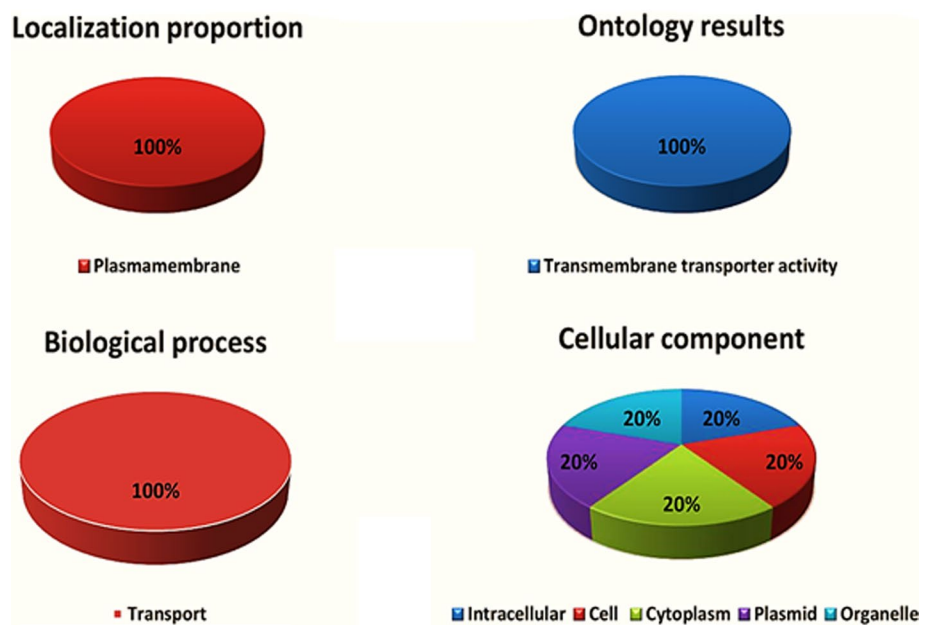


Fig. 9 Protein interaction network of CsNHX proteins

Fig. 10 Subcellular localization, biological process, cellular component, and gene ontology analysis of KEA genes



Gossypium hirsutum, *Glycine max*, *Gossypium raimondii*, *Glycine soja*, *Helianthus annuus* L., *Helianthus argophyllus* Torr. & A.Gray, *Hevea brasiliensis*, *Helianthus ciliaris* DC., *Helianthus exilis* A.Gray, *Helianthus paradoxus* Heiser, *Helianthus tuberosus* L., *Hordeum vulgare*, *Lotus japonicus*, *Malus domestica*, *Manihot esculenta* Crantz, *Medicago truncatula*, *Nicotiana tabacum*, *Oryza sativa*,

Phaseolus vulgaris L., *Picea abies* (L.) H.Karst., *Pinus densata*, *Pinus taeda*, *Populus euphratica* Oliv., *Populus trichocarpa*, *Ricinus communis* L., *Rehmannia glutinosa*, *Saccharum* sp., *Solanum tuberosum*, *Sorghum bicolor*, *Solanum lycopersicum*, *Saccharum officinarum* L., *Salvia sclarea* L., *Triticum aestivum*, *Theobroma cacao*, *Vigna unguiculata* (L.) Walp., *Vitis vinifera*, *Zea mays*). The gene least targeted

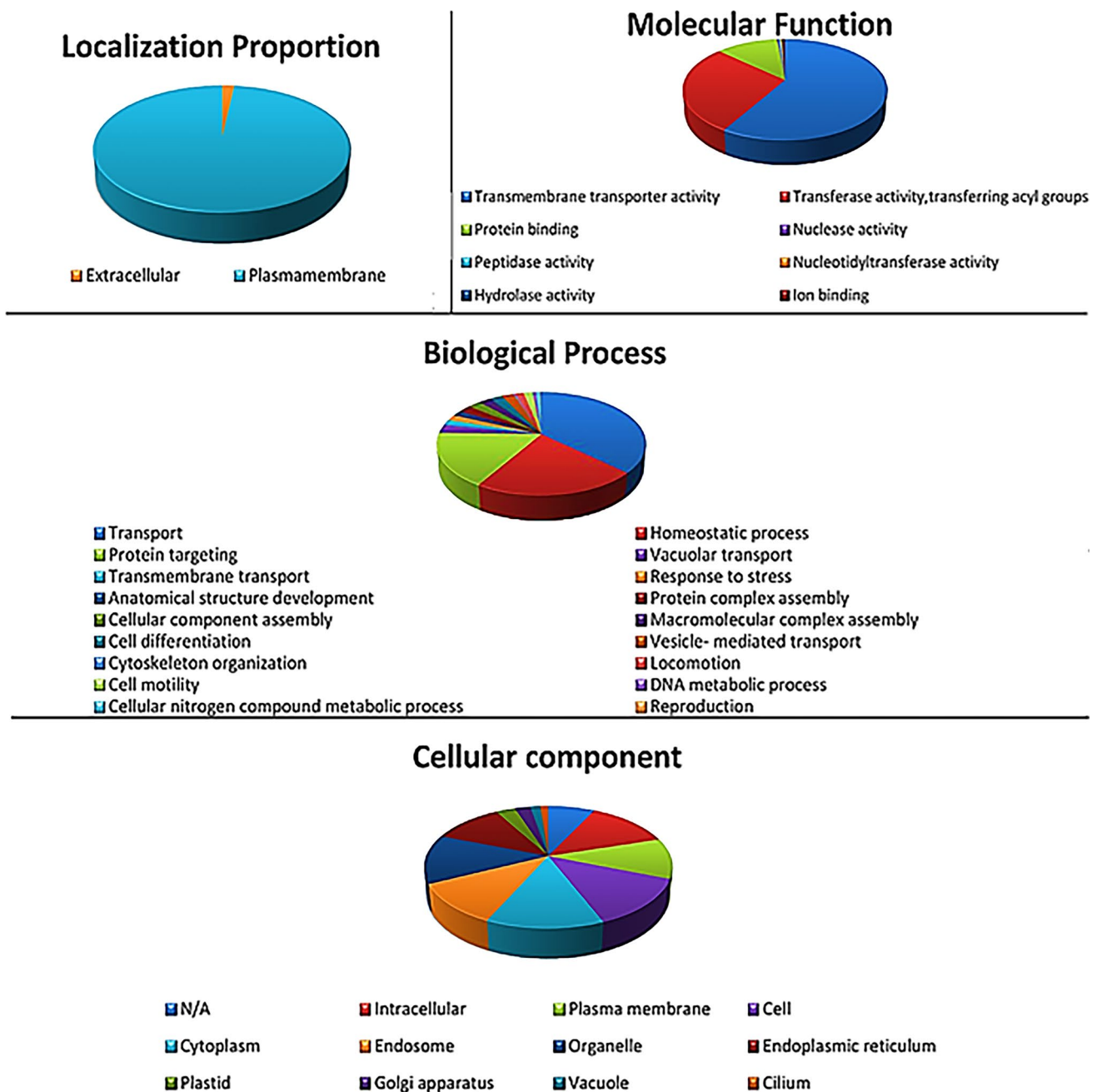


Fig. 11 Subcellular localization, biological process, cellular component, and gene ontology analysis of *NHX* genes

by miRNA is *CsNHX9* (27 miRNAs). The genes most targeted by miRNA are *CsNHX5* (517 miRNAs) and *CsNHX14* (517 miRNAs). The other targeted genes and miRNA counts are as follows: *CsNHX13* (514 miRNAs), *CsNHX26* (391 miRNAs), *CsNHX20* (346 miRNAs), *CsNHX18* (336 miRNAs), *CsNHX19* (331 miRNAs), *CsNHX17* (328 miRNAs), *CsNHX3* (319 miRNAs), *CsNHX54* (281 miRNAs), *CsNHX21* (270 miRNAs), *CsNHX11* (242 miRNAs), *CsNHX51* (239 miRNAs), *CsNHX45* (233 miRNAs), *CsNHX52* (223 miRNAs), *CsNHX28* (222 miRNAs),

CsNHX27 (220 miRNAs), *CsNHX24* (209 miRNAs), *CsNHX23* (205 miRNAs), *CsNHX25* (205 miRNAs), *CsNHX22* (202 miRNAs), *CsNHX53* (198 miRNAs), *CsNHX7* (190 miRNAs), *CsNHX55* (178 miRNAs), *CsNHX2* (176 miRNAs), *CsNHX12* (175 miRNAs), *CsNHX41* (175 miRNAs), *CsNHX42* (166 miRNAs), *CsNHX6* (161 miRNAs), *CsNHX30* (150 miRNAs), *CsNHX44* (149 miRNAs), *CsNHX29* (148 miRNAs), *CsNHX16* (145 miRNAs), *CsNHX50* (144 miRNAs), *CsNHX46* (139 miRNAs), *CsNHX43* (137 miRNAs), *CsNHX49*

Table 1 The selected primers for PCR and their properties

Primers	Features
> CsKEA1:f_322-342 322-> 342 5'-ctttggacaaactagctcgat	Length=21 A=6,0 G=4,0 T=6,0 C=5,0 CG=42,9% Linguistic complexity=95% Primer's PCR efficiency=95% Tm=57,0 °C Amplicon size: 1229 bp Ta=65 °C
> CsKEA1:r_1531-1550 1531<-1550 5'-caagtacatttgccactct	Length=20 A=5,0 G=3,0 T=6,0 C=6,0 CG=45,0% Linguistic complexity=95% Primer's PCR efficiency=95% Tm=58,0 °C Amplicon size: 1229 bp Ta=65 °C
> CsNHX5:f_429-448 429-> 448 5'-gctggaacttttagtcccaat	Length=20 A=5,0 G=4,0 T=6,0 C=5,0 CG=45,0% Linguistic complexity=97% Primer's PCR efficiency=97% Tm=54,7 °C Amplicon size: 816 bp Ta=65 °C
> CsNHX5:r_1226-1244 1226<-1244 5'-aggaaatgcttcgtcaacc	Length=19 A=6,0 G=4,0 T=4,0 C=5,0 CG=47,4% Linguistic complexity=97% Primer's PCR efficiency=97% Tm=54,8 °C Amplicon size: 816 bp Ta=65 °C
> CsNHX14:f_429-448 429-> 448 5'-gctggaacttttagtcccaat	Length=20 A=5,0 G=4,0 T=6,0 C=5,0 CG=45,0% Linguistic complexity=97% Primer's PCR efficiency=97% Tm=54,7 °C Amplicon size: 816 bp Ta=65 °C
> CsNHX14:r_1226-1244 1226<-1244 5'-aggaaatgcttcgtcaacc	Length=19 A=6,0 G=4,0 T=4,0 C=5,0 CG=47,4% Linguistic complexity=97% Primer's PCR efficiency=97% Tm=54,8 °C Amplicon size: 816 bp Ta=65 °C
> CsNHX18:f_450-468 450-> 468 5'-aggatattcctcacgctgt	Length=19 A=4,0 G=4,0 T=6,0 C=5,0 CG=47,4% Linguistic complexity=97% Primer's PCR efficiency=97% Tm=57,5 °C Amplicon size: 1766 bp Ta=64 °C
> CsNHX18:r_2196-2215 2196<-2215 5'-ctcgagcttggtcagaaata	Length=20 A=6,0 G=5,0 T=5,0 C=4,0 CG=45,0% Linguistic complexity=95% Primer's PCR efficiency=95% Tm=57,5 °C Amplicon size: 1766 bp Ta=64 °C

(126 miRNAs), *CsNHX34* (125 miRNAs), *CsNHX40* (125 miRNAs), *CsNHX8* (121 miRNAs), *CsNHX31* (117 miRNAs), *CsNHX47* (114 miRNAs), *CsNHX1* (106 miRNAs), *CsNHX32* (104 miRNAs), *CsNHX39* (101 miRNAs), *CsNHX10* (102 miRNAs), *CsNHX15* (102 miRNAs), *CsNHX33* (101 miRNAs), *CsNHX35* (100 miRNAs), *CsNHX36* (99 miRNAs), *CsNHX37* (83 miRNAs), *CsNHX38* (81 miRNAs), *CsNHX48* (81 miRNAs) and *CsNHX4* (37 miRNAs) (Supplementary File 5).

miR156 is the most abundant miRNA linked with the stress response in *CsNHX*, while miR395 is the most abundant miRNA in *CsKEA* (In *CsNHX*, miR395 is also in the second position). Furthermore, *CsNHX* also has stress-associated miR159, miR160, miR164, miR165, miR166, miR167, miR168, miR169, miR170, miR171, miR172, miR319, miR390, miR393, miR394, miR396, miR397, miR398, miR408, miR474, miR827, miR1432, miR1450, and miR5381. *CsKEA* includes all miRNAs except miR160, miR164, miR165, miR170, miR393, miR394, miR474, miR1432, miR1450 and miR5381. miR162 is detected in *CsKEA* but not *CsNHX*. In addition, miR5185 is a family of miRNAs that is abundant in both *CsKEA* and *CsNHX*.

According to the analysis, miR156 mostly targets *CsNHX5*, while miR395 targets *CsNHX11* (Supplementary File 5).

In silico PCR primer search – Polymerase chain reaction (PCR) is the basis of molecular biology, and primer design is a critical step in all PCR methods (Kalendar et al. 2011). For this reason, FastPCR software was used to search for primers. When the results obtained were evaluated, *CsKEA1*, *CsNHX5*, *CsNHX14*, and *CsNHX18* were selected as genes that could be used for PCR (Table 1).

4 Discussion

The impact of abiotic stressors like water scarcity and salt stress on agriculture is worsened by climate change. Therefore, there is an increasing interest in biotechnology-based research into the improvement of abiotic stress tolerance (Yang et al. 2021). Sustainable genetic improvements in crops can be achieved through functional genomics and gene-modification techniques, including the CRISPR/Cas9 system for characterizing stress-related genes, along with

the use of microbial transgenes to enhance crop tolerance and improve yield and quality (Kausar et al. 2022). Developing salt-tolerant crops using transgenic technology are vital for their resilience in challenging environments. Prior to creating transgenic crops, a critical step involves identifying candidate genes that play key roles in various metabolic pathways, such as osmolyte synthesis and selective ion uptake to maintain ion homeostasis. For this purpose, various *NHX* genes can be utilized to obtain stress-resistant transgenic plants (Yara 2019). In this study, the roles of *KEA* and *NHX* genes in response to abiotic stress were investigated, and genes with roles in stress responses were explored as potential candidates for obtaining abiotic stress-resistant transgenic plants.

In studies on the *KEA* genes, 6 *KEAs* were determined in *Arabidopsis thaliana*, 10 *KEAs* in *Raphanus sativus* L., 13 *KEAs* in *Brassica rapa*, 12 *KEAs* in *Pyrus bretschneideri* Rehder, 7 *KEAs* in *Malus domestica*, 5 *KEAs* in *Prunus persica* (L.) Batsch, 5 *KEAs* in *Fragaria vesca* L., 4 *KEAs* in *Prunus mume* (Sieb.) Sieb. et Zucc., 4 *KEAs* in *Vitis vinifera*, 4 *KEAs* in *Oryza sativa*, 6 *KEAs* in *Zea mays*, 4 *KEAs* in *Selaginella* P. Beauv., 4 *KEAs* in *Ostreococcus* C.Courties & M.-J.Chrétiennot-Dinet, 3 *KEAs* in *Chlamydomonas reinhardtii*, 7 *KEAs* in *Physcomitrella patens* (Hedw.) Bruch & Schimp., 7 *KEAs* in *Populus trichocarpa*, 7 *KEAs* in *Gossypium raimondii*, 6 *KEAs* in *Phyllostachys edulis* (Carrière) J. Houz., 6 *KEAs* in *Vigna radiata* (L.) R. Wilczek, 6 *KEAs* in *Cajanus cajan* (L.) Mill sp. and 6 *KEAs* in *Cicer arietinum* L. (Wang et al. 2020; Wu et al. 2020; Siddique et al. 2021; Azeem et al. 2022a, 2022b). *Sorghum bicolor* was reported to have 4 *KEAs* in one study (Wang et al. 2020) while it was reported to contain 2 *KEAs* in another study (Anil Kumar et al. 2022). In this research, two *KEAs* were found in *Citrus sinensis* (L.) Osbeck. When the mentioned articles about *KEA* genes were examined, it was determined that the number of *KEA* genes in *C. sinensis* was lower than in other articles, except for a few. In the studies on the presence of *NHX* genes, it was revealed that there are 16 *NHX* genes in *Oryza sativa*, 13 *NHXs* in *Zea mays*, 10 *NHXs* in *Punica granatum* L., 23 *NHXs* in *Gossypium hirsutum*, 24 *NHXs* in *Gossypium barbadense* L., 12 *NHXs* in *Gossypium raimondii*, 12 *NHXs* in *Gossypium arboreum* L., 9 *NHXs* in *Camellia sinensis* (L.) Kuntze, 5 *NHXs* in *Beta vulgaris* L., 7 *NHXs* in *Lonicera japonica* Thunb., 8 *NHXs* in *Canavalia rosea* (Sw.) DC., 14 *NHXs* in *Pyrus bretschneideri*, 12 *NHXs* in *Malus domestica*, 7 *NHXs* in *Prunus mume*, 6 *NHXs* in *Fragaria vesca*, 6 *NHXs* in *Prunus persica*, 8 *NHXs* in *Arabidopsis thaliana*, 9 *NHXs* in *Raphanus sativus*, 11 *NHXs* in *Brassica rapa*, 8 *NHXs* in *Vitis vinifera*, 7 *NHXs* in *Sorghum bicolor*, 7 *NHXs* in *Selaginella*, 6 *NHXs* in *Ostreococcus*, 9 *NHXs* in *Chlamydomonas reinhardtii*, 10 *NHXs* in *Physcomitrella patens* and 8 *NHX* genes in *Populus trichocarpa* (Zhou et al. 2016; Wu et al. 2019; Fu et al. 2020; Wang et al. 2020; Dong et al.

2021; Khare et al. 2021; Kong et al. 2021; Paul et al. 2021; Huang et al. 2022; Pu et al. 2022). Contrary to these, fifty-five *CsNHX* genes were identified in this article.

When we examined the subcellular localization of *KEA* and *NHX* proteins, it was determined that all proteins except *NHX9* were found only in the plasma membrane, but *NHX9* was also located in the extracellular. It has been stated that the *NHX* proteins of *Gossypium barbadense* and *G. hirsutum* are localized in the plasma membrane, tonoplast, and endomembrane (Akram et al. 2020). According to previous research (Dong et al. 2021), the majority of the *NHX* proteins of *Punica granatum* are found in the vacuole and a smaller amount in the cell membrane. According to a separate study (Wu et al. 2019), *NHX* proteins belonging to *Beta vulgaris* are localized in the vacuole, plasma membrane, and endosome. When the roles of *KEA* and *NHX* proteins in biological processes were investigated, it was observed that *KEA* proteins were involved in transport. It has been determined that *NHX* proteins have roles in transport, protein targeting, transmembrane transport, anatomical structure development, cellular component assembly, cell differentiation, cytoskeleton organization, cell motility, cellular nitrogen compound metabolic process, homeostatic process, vacuolar transport, response to stress, protein complex assembly, macromolecular complex assembly, vesicle-mediated transport, locomotion, DNA metabolic process, and reproduction. It was defined that nine of the fifty *NHX* proteins (*CsNHX5*, *CsNHX11*, *CsNHX12*, *CsNHX17*, *CsNHX27*, *CsNHX28*, *CsNHX47*, *CsNHX48*, and *CsNHX55*) play a role in the stress response. In support of this study, it has been reported that *Arabidopsis thaliana*, *Beta vulgaris*, *Glycine max*, *Gossypium barbadense*, *Gossypium hirsutum*, *Hordeum vulgare*, *Oryza sativa*, *Solanum lycopersicum*, *Sorghum bicolor*, *Triticum aestivum*, and *Zea mays* *NHX* genes respond to salinity stress (Akram et al. 2020). It has been observed that the relative expression levels in the leaves of *Punica granatum* increase as the amount of Na^+ increases under low and moderate salt stress, and it is also thought that the *NHX* genes of *P. granatum* may mitigate the deleterious effects of Na^+ (Dong et al. 2021).

When the data gathered for the article were analyzed, it was found that phylogenetic analysis, gene structure analysis, and conserved protein motif analysis supported one another. As a result of the phylogenetic analysis, three distinct clusters were observed in the phylogenetic tree of *NHX* proteins. When examining the lower branches, it was established that *CsNHX55* belonged to a distinct branch, despite the fact that Cluster II contained all the genes implicated in the stress response. The gene structure analysis clearly confirmed this distinction. When we compared the results of the motif analysis of these stress-responding genes, it was determined that the motifs of *CsNHX27* and *CsNHX28* were similar to each other, while the motifs of *CsNHX11*, *CsNHX12*, *CsNHX17*,

CsNHX47, and CsNHX48 were similar to each other. The motifs of CsNHX5 and CsNHX55 differ from the others. CsNHX5 contains only the 1st and 7th motifs, whereas the CsNHX55 protein contains only the 3rd, 5th, 6th, and 10th motifs. A comprehensive investigation of the phylogenetic tree demonstrates that *CsNHX5* has already diverged from other stress response genes. Exon and intron structural diversity play a crucial role in the evolution of gene families (Wu et al. 2019). Introns are intervening DNA sequences widely distributed throughout gene coding regions in eukaryotes and are involved in alternative splicing. It has been reported that these sequences can significantly affect gene expression by affecting transcription, translation, and mRNA degradation (Hernandez-Garcia and Finer 2014; Siddique et al. 2021). Analysis of the structure of the *KEA* and *NHX* genes of *Citrus sinensis* (L.) Osbeck revealed the presence of introns in all genes except *CsNHX4*. Similarly, it was stated that only the *OsNHX12* gene in Indica rice lacks introns (Khare et al. 2021). Furthermore, in present study, it was determined that the *CsNHX4* consisted of only 2 motifs. The gene with the maximum introns and exons was observed to be *CsNHX54*. *CsNHX52* was analyzed as the second gene containing these regions the most. When the phylogenetic tree was looked at, these two genes were found to be part of Cluster II. However, when the distribution of genes on chromosomes was studied, it was observed that, with the exception of *CsNHX9*, all genes in the first branch of Cluster II (*CsNHX52*, *CsNHX53*, *CsNHX54*, and *CsNHX55*) were scaffolds. Interestingly, *CsNHX9* also has the following features: the subcellular localization of just the CsNHX9 protein is extracellular. The intron and exon sections are extremely short. This gene is the least targeted by miRNA. Another important finding is that although CsNHX55 is scaffolded, it is central to protein–protein interactions, suggesting that the CsNHX55 protein has an important role.

Segmental and tandem duplications play an important role in the evolution of species (He et al. 2022). Expansion in gene families is the result of gene duplications (Wang et al. 2020). Gene duplications increase functional diversity, which is essential for environmental change adaptation (Paul et al. 2021). In the *NHX* genes of *C. sinensis*, tandem and segmental duplications were identified. It has been determined that tandem duplications are mostly on the 7th chromosome. This is expected because, as a result of the chromosome distribution analysis, the chromosome with the highest number of genes (*CsNHX34*, *CsNHX35*, *CsNHX38*, *CsNHX39*, *CsNHX40*, *CsNHX41*, *CsNHX42*, *CsNHX43*, *CsNHX44*, *CsNHX45*) was determined to be the 7th chromosome. According to the report, tandem and segmental duplications were identified in the *NHX* genes of *Vitis vinifera*; however, tandem duplication did not occur in the *NHX* genes of *Camellia sinensis* (Ayadi et al. 2020; Paul et al. 2021). When segmental duplications were analyzed in

the present study, there was segmental duplication between *CsNHX1* and several genes (*CsNHX6*, *CsNHX7*, *CsNHX8*, *CsNHX10*, *CsNHX13*, *CsNHX14*, *CsNHX15*, *CsNHX16*, *CsNHX23*, *CsNHX24*, *CsNHX25*, *CsNHX26*, *CsNHX29*, *CsNHX30*, *CsNHX31*, *CsNHX32*, *CsNHX33*, *CsNHX34*, *CsNHX39*, *CsNHX40*, *CsNHX41*, *CsNHX42*, *CsNHX43*, *CsNHX44*, *CsNHX45*, *CsNHX46*). Interestingly, it has been shown that genes involved in the stress response, such as *CsNHX11*, *CsNHX12*, exhibit segmental duplication with each other but not with other genes (Supplementary File 3).

In silico genome-wide analysis of cis-elements is important for a better understanding of gene regulation (Hernandez-Garcia and Finer 2014). Cis-acting elements are important molecular structures in the transcriptional regulation of genes that control biological processes such as abiotic stress responses and understanding the regulatory genes involved in the stages of stress responses is possible by analyzing the functional analyses of cis-acting elements (Yamaguchi-Shinozaki and Shinozaki 2005). In this study, several cis-acting elements, such as LTR, MBS, and TC-rich repeats, were identified in the promoter regions of the *CsKEA* and *CsNHX* genes. LTR, the motif associated with low-temperature stress, was observed only in *CsKEA1*, whereas MBS, the motif related to drought, was observed only in *CsKEA2*. This suggests that *CsKEA1* may have a role in low-temperature stress and *CsKEA2* in drought stress. As a result of the analysis, it was observed that *CsNHX5* and *CsNHX48*, which are genes that have a function in the stress response, do not contain any of the LTR, MBS, and TC-rich repeats. In addition, only TC-rich repeats acting elements were detected in the promoter regions of *CsNHX11*, *CsNHX27*, *CsNHX28* and *CsNHX55*, only MBS in *CsNHX12*, and only LTR in *CsNHX47*. On the other hand, *CsNHX17* has all the indicated motifs. *CsNHX12* is a gene associated with the drought response, while *CsNHX17* may be implicated in both drought and low-temperature stress, according to the data. Plants respond to environmental stimuli, and plant hormones such as abscisic acid (ABA), salicylic acid (SA), and jasmonates (JA) are involved in the regulation of stress responses (Arbona et al. 2017). ABA is a plant hormone produced under abiotic stress conditions such as drought and high salinity, which increases the activation of many stress-related genes and the tolerance of plants to stress (Nakashima et al. 2012). ABRE, an ABA response element, has been reported to regulate the dehydration and salinity responses in *Arabidopsis* and rice (Hernandez-Garcia and Finer 2014). In this study, as a result of the analysis, *CsNHX48* was determined to be the gene with the most ABRE. *CsNHX48* is already a gene involved in stress response, which is one of the evidences that functional gene ontology and cis-acting element analysis complement each other. On the other hand, it was found that the gene with the highest number of CGTCA-motif and

TGACG-motifs involved in the methyl jasmonate response was *CsNHX11*, while SARE, a motif associated with the salicylic acid response, was found only in *CsNHX27* and *CsNHX28*. *CsNHX11*, *CsNHX27*, and *CsNHX28* are genes like *CsNHX48*, which were determined as a result of the analysis to be involved in the stress response. Additionally, *CsKEA* genes contain motifs linked to salicylic acid, methyl jasmonate, and abscisic acid responses. This indicates that *CsKEA* genes may react to abiotic stresses.

Secondary metabolites are compounds that are produced in small amounts by plants yet have numerous essential therapeutic qualities, such as antioxidant and antibacterial activities. It has been reported that abiotic stress stimulates the expression of flavonoid biosynthesis genes and the accumulation of flavonoids, one of the secondary metabolites (Arbona et al. 2017). It is observed that *CsNHX2*, *CsNHX7*, *CsNHX8*, *CsNHX24*, *CsNHX25*, and *CsNHX34* contain the MYB binding site involved in the regulation of flavonoid biosynthetic genes. Flavonoids have effects such as antioxidant, antibacterial, anticancer, cardioprotective, anti-inflammatory, promoting immune system (Tungmunthum et al. 2018). The *CsNHX2*, *CsNHX7*, *CsNHX8*, *CsNHX24*, *CsNHX25*, and *CsNHX34* identified in *C. sinensis* may also contribute to the medicinal properties of this plant.

Computational techniques have been instrumental in the detection and validation of miRNAs, particularly those challenging to clone due to their low expression levels, across various plant species. These techniques use criteria such as sequence conservation, secondary structure features, and target-guided strategies to predict new miRNAs, facilitating comprehensive studies in plants, such as *Arabidopsis*, rice, and wheat, even in the absence of complete genome annotations (Tripathi et al. 2015). MicroRNAs (miRNAs) are short, non-coding regulatory RNAs consisting of 20–24 nucleotides that play crucial roles in plants by targeting mRNAs (Lu and Huang 2008). Numerous miRNAs in plants play an essential role in the development of abiotic stress resistance (Sunkar et al. 2007). It is believed that plants exposed to abiotic stress might develop stress tolerance by regulating some of their miRNAs and generating new miRNAs in order to survive and carry out their physiological functions (Zhang et al. 2022a, b). It has been noticed that miRNAs such as miR168, miR171, and miR396 are responsive to drought, cold, and salinity stress in *Arabidopsis* (Khraiweh et al. 2012). In another study (Sunkar et al. 2012), miR156, miR159, miR160, miR162, miR165, miR166, miR167, miR168, miR169, miR170, miR171, miR172, miR319, miR390, miR393, miR395, miR396, miR397, miR398 and miR408 responses to drought stress were reported. Moreover, a report indicates that miR156, miR159, miR160, miR162, miR167, miR168, miR169, miR170, miR171, miR319, miR393, miR395, miR396, miR397, miR398 and miR408 respond to salinity and cold stress, whereas miR165,

miR166, and miR172 respond to cold stress. In addition, data on the involvement of miR156, miR159, miR160, miR165, miR166, miR168, miR169, miR170, miR171, miR172, and miR393 in the heat response, and miR159, miR160, miR167, miR169, miR319, miR393, miR397, and miR398 in the ABA response were given in the same article. Another stress in the same study was oxidative stress, and it was observed that miR169, miR397, and miR398 induced an oxidative stress response. The miRNAs that respond to UVB are listed as follows: miR156, miR159, miR160, miR165, miR166, miR167, miR168, miR170, miR171, miR172, miR393, and miR398. miRNAs that respond to hypoxia stress were identified in the same article as miR156, miR159, miR160, miR162, miR165, miR166, miR167, miR168, miR169, miR170, miR171, miR172, miR390, miR395 and miR396. According to the results of a different study (Shukla et al. 2008), miR159 may play a significant function in the hormonal and abiotic stress signaling networks. Moreover, miR474 and miR528 are important in drought response (Ding et al. 2013). In addition, another study (Tiwari and Rajam 2022) demonstrated that miR528 is linked to salt tolerance. It is stated that miR394 modulates plant hormones, including ABA, JA, and SA, and that overexpression of this miRNA enhances drought tolerance in transgenic *Arabidopsis* (Zhang et al. 2022a, b). Apart from these, it was concluded that miR529, miR845, miR851, miR854, miR896, miR901, miR903, miR1030, miR1035, miR1050, miR1088, miR1125, and miR1126 were up- or down-regulated during the drought (Guleria et al. 2011). In this study, it was determined that miR160, miR165, miR170, miR393, miR394, miR474, miR528, miR845, miR851, miR854, miR896, miR901, miR903, miR1030, miR1035, miR1050, miR1088, miR1125, and miR1126, which were mentioned in previous studies, do not target *CsKEA* genes. In addition, it was found that miR159, miR166, miR169, miR172, miR395, miR396, miR398, and miR529 target both genes (*CsKEA1* and *CsKEA2*); miR162, miR390, and miR408 target only *CsKEA1*; and miR156, miR167, miR168, miR171, miR319, and miR397 target only *CsKEA2*. When miRNAs given in previous studies were examined in terms of *CsNHX* genes, it was observed that miR162, miR896, miR901, miR903, miR1030, miR1035, miR1050, miR1088, and miR1126 did not target any genes. The genes targeted by other miRNAs are given in Supplementary File 5. It was concluded that miR168, miR170, miR390, miR528 and miR1125 do not target the genes found to have a role in the stress response in this article. When examined in terms of these genes involved in the stress response, it was determined that miR160 targets only the *CsNHX11* gene, miR166 only targets *CsNHX55*, miR394 targets only *CsNHX48* and miR851 targets only *CsNHX5*. At least two of the genes involved in stress have also been seen to be targeted by other miRNAs. miR394 is involved in the regulation of plant hormones such as

salicylic acid and abscisic acid. In this article, ABRE was found in the highest number in *CsNHX48*, one of the genes involved in the stress response. Abscisic acid is known to be an important hormone in abiotic stress. As previously mentioned, miR394 only targeted *CsNHX48*. These findings also support the hypothesis that *CsNHX48* may play a role in the abiotic stress response. According to the obtained data, miRNA analysis, cis-acting element analysis, and gene ontology analysis are complementary. When all the information is put together, it seems likely that *CsNHX* genes and miRNAs that target these genes may function in Na⁺ transport in *Citrus sinensis* (L.) Osbeck. Another study (Khare et al. 2021) on *NHX* genes supports this idea.

The PCR technique, while being a significant tool in molecular biology, relies on the design of efficient primers for successful outcomes. To achieve this, various web tools were employed, and data from FastPCR software, along with data from other laboratories, were collected and analyzed to predict the quality of oligonucleotides more accurately for PCR efficiency. By adjusting the parameters for oligonucleotide design based on this data, the success rate of PCR has been enhanced (Kalender, 2011). Using software for PCR offers significant time and cost savings compared to traditional laboratory-based methods. Additionally, numerous external factors can introduce errors in the laboratory process. Primers for stress-resistant genes obtained through software can be tested in the laboratory, resulting in a quick and effective procedure. Primers are small pieces of DNA and RNA from which DNA synthesis is initiated, and the primer length varies between 18 and 30 nucleotides. In addition, a good primer should contain 45–60% GC (guanine and cytosine) (Topu et al. 2019). In this study, primers were identified using the FastPCR software, and those that met the specified criteria were selected. The *CsKEA1*, *CsNHX5*, *CsNHX14*, and *CsNHX18* meet the mentioned criteria. Reviewing the analyses, it was found that *CsNHX14* has MBS, which is a motif related to drought, and TC-rich repeats, which are associated with defense and stress. It has also been observed that this gene is one of the most targeted genes by miRNAs. Aside from these, protein–protein interactions reveal that the *CsNHX17* and *CsNHX55* proteins, which have been linked to the stress response, interact with *CsNHX14*. On the other hand, the promoter region of *CsNHX18* had LTR, a low-temperature motif, a wound-related motif called WUN, and MBS. One of the genes most targeted by miRNA is *CsNHX5*. miR156 mostly targets *CsNHX5*. miR156 is a stress-related miRNA such as drought, salinity, cold, heat, hypoxia, and UVB. According to gene function analysis, *CsNHX5* has been found to be involved in the stress response. The obtained data support each other. Additionally, miR474 targets *CsNHX5* and is related to drought. Likewise, another drought-related miRNA, miR851, targets only *CsNHX5*. In terms of the

quantity of cis-acting elements, this gene ranks second. The gene with the most TATA-box and CAAT-box is *CsNHX5*. When the phylogenetic tree is investigated, *CsNHX5* is revealed to be distinct.

In conclusion, the comprehensive bioinformatics analysis conducted in this study has shed light on the critical roles played by *CsKEA* and *CsNHX* genes in various plant physiological processes, with a particular emphasis on the *NHX* genes' involvement in stress responses. When the data obtained from all the analyses is supported by experimental studies, it will become even more reliable. The results strongly indicate that *CsNHX* genes have the potential to confer protection against a range of abiotic stresses, including salinity, drought, and temperature fluctuations. Notably, *CsNHX5* has emerged as a promising candidate for further investigation, suggesting its potential utility in transgenic plant production for enhanced stress tolerance. This research provides valuable insights into the molecular mechanisms governing stress responses in plants and opens up exciting avenues for future studies in this field.

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Declarations

Conflict of interest The author declares no conflict of interest.

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